AN AI-ASSISTED DIGITAL TAILOR SYSTEM FOR CUSTOM CLOTHING FIT AND FABRIC WASTE REDUCTION USING DESIGN IMAGE ANALYSIS

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by

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Table of Contents

1.Introduction 3
1.1 Background of the Study 3
1.2 Motivation 3
1.3 Study Objectives 4
2.Literature Review and Research Gap 4
2.1 Overview of Existing Studies 4
2.2 Identification of Research Gap 5
3.Research Problem and Questions 6
3.1 Main Research Problem 6
3.2 Specific Research Questions 6
4.Research Strategy and Methodology 7
4.1 Research Strategy 7
4.2 Research Methodology Framework 7
4.3 Data Collection Methods 8
4.3.1 Primary Data 8
4.3.2 Secondary Data 8
4.4 Data Preprocessing / Preparation 9
4.5 Model Development / Implementation 9
4.6 Evaluation 10
4.6.1 Fit Accuracy 10
4.6.2 Fabric Waste Reduction 10
4.6.3 Usability and User Satisfaction 10
4.7 Ethical considerations 10
4.8 Limitations of the study 11
5.Timeline 11
6.References 13

1.Introduction

1.1 Background of the Study

The clothing and fashion industry is growing very fast around the world, but it also creates a large amount of fabric waste and environmental damage. In ordinary tailoring and custom clothing, the amount of fabric needed is usually guessed from manual body measurements and the tailor's experience. To reduce the risk of getting the size wrong, many tailors cut extra cloth. This "safe" approach wastes a lot of material. Online clothing shops face a related problem. They depend on standard size charts, which do not capture the real differences in people's body shapes as a result, most customers receive clothes that do not fit well, and there is unnecessary usage of fabric

In the past few years, there have been advancements in AI, machine learning, and computer vision. Computer vision can analyze an image, spot a pattern, and plan based on that pattern found in the image. In fashion, they are already used in tools like virtual try-on systems, automatic size suggestions, and design analysis. The main goal of these tools is to improve the shopping experience and reduce size-related problems in online buying.

Even with these advances, most current systems focus only on how a garment looks on the body or which size a person should pick. They rarely try to estimate the exact amount of fabric needed for a specific person and design. In addition, many digital tailoring tools do not use the image of clothing design as an input for creating personalized measurements or for calculating fabric requirements. There is still a clear gap: we lack integrated systems that link design-image analysis, custom sizing, and fabric optimization. Filling this gap is important not only for better-fitting custom garments but also for supporting sustainable fashion by cutting down material waste.

1.2 Motivation

This study is supported by practical, academic, and social reasons. In practice, many tailors, fashion designers, and customers still face problems with clothes that do not fit properly and with fabric being wasted. Using more cloth than needed raises production costs and adds to environmental problems because more textile waste is produced.

From the academic side, modern data-driven computer methods are now widely used to solve everyday problems, which creates a chance to test new ideas in fashion technology. As a software engineering undergraduate, the researcher wants to use knowledge of intelligent computer techniques, image handling, and web-based system development to work on a real problem in the clothing field.

Additionally, given the increasing focus on sustainability and the judicious usage of resources, the issue has also been driven by the need for responsible environmental practices by the garment industry. An AI-assisted digital tailoring system that seeks to improve garment fit accuracy and reduce fabric waste is an application that contributes fully to sustainable development principles.

1.3 Study Objectives

Primary Objective

To design and develop an AI-assisted digital tailor system that generates personalized clothing size charts and accurately estimates fabric requirements using clothing design image analysis, with the aim of improving garment fit and reducing fabric waste.

Secondary Objectives

- To analyze uploaded clothing design images and extract design-related features that influence garment sizing and material usage.
- To generate customized size charts based on user-provided body details using AI-driven prediction techniques.
- To calculate the exact quantity of fabric required for custom clothing production based on the generated size and selected design.
- To compare traditional fabric estimation methods with the proposed AI-based approach in terms of material usage and waste reduction.
- To develop a web-based prototype that demonstrates the functionality of the proposed digital tailor system.
- To evaluate the effectiveness of the system in improving fit accuracy and reducing fabric waste.

2.Literature Review and Research Gap

2.1 Overview of Existing Studies

In recent years, there has also been a rising trend of research concerning the integration of artificial intelligence (AI) and digital technology applied to the fashion and clothing industry. The literature records the emergence of AI-based methodologies that have been invented to enhance the fit of garments, provide automated size suggestions, and ensure that clothing production is sustainable. Some studies specifically concentrate on AI-based size estimation and personalized fitting solutions. For example, machine learning models have been used to predict clothing sizes using anthropometric data obtained from body measurements or 3D body scans. These approaches demonstrate improved accuracy compared to traditional standard size charts and reduce fitting errors in online clothing platforms [1],[2],[3]. However, these systems primarily concentrate on size recommendation and do not consider material usage or fabric optimization.

Another significant research direction involves computer vision and image-based garment analysis. Image-based virtual try-on systems use deep learning techniques to visualize how garments fit on human models or users [4],[5] . Such systems enhance the online shopping

experience by reducing uncertainty related to garment appearance and fit. Although virtual try-on tools are good for showing how clothes might look on a person, they usually do not create proper size charts or work out how much fabric is needed to make the actual garment.

Research has also investigated the use of AI in sustainable fashion and textile waste reduction. Studies highlight that excessive fabric cutting and inefficient material planning are major contributors to textile waste [6]. There have been proposals based on AI-related optimization methods that can improve production planning and minimize waste in production settings [6], [7]. Even if these studies can relate to sustainability, they often focus on supply chain optimization, recycling, and mass production settings instead of tailored production.

Some researchers have explored fabric property prediction and material analysis using AI. These studies apply machine learning models to predict fabric characteristics such as thickness, drape, and elasticity [8]. Although useful for fabric selection, such approaches are rarely integrated with personalized size generation or design-specific fabric quantity calculation.

A limited number of studies have attempted to combine digital tailoring concepts with AI-based systems. These systems aim to automate measurement processes and improve efficiency in custom garment production [9]. While many digital tailoring systems today rely on manual entry of garment measurements and a pre-set rules system, they do not address the intelligent analysis of design images and the calculation of material quantities. Together, the current literature indicates great potential for the use of AI to improve clothing fit, personalization, and sustainability in fashion. Yet, most literature attempts to address the aforementioned perspectives separately rather than being an integrated system for design image analysis, individualized size development, and material cutting waste reduction.

2.2 Identification of Research Gap

Based on the review of existing literature, several key research gaps can be identified.

Firstly, while many studies focus on AI-based clothing size recommendation, they largely rely on standard body measurements or 3D scanning data and do not utilize clothing design images as an input for size generation [1], [2], [7]. The influence of garment design features on sizing and material requirements remains underexplored.

Secondly, current image-based fashion systems, such as virtual try-on applications, primarily aim at visualization and user experience enhancement [4], [5]. These systems do not generate practical outputs such as customized size charts or fabric quantity calculations, which are essential for real-world tailoring and garment production.

Thirdly, although sustainability and fabric waste reduction are widely discussed in the literature, most AI-based solutions address waste reduction at the manufacturing or recycling stage [6]. There

is a lack of research focusing on fabric waste minimization at the individual custom tailoring level, where inaccurate estimation and overcutting frequently occur.

Finally, existing digital tailoring systems do not sufficiently integrate AI-driven decision-making, personalized fitting, and material optimization into a single web-based platform [9]. The absence of such integrated systems highlights a clear research gap.

To address these limitations, this study proposes an AI-assisted digital tailor system that analyzes clothing design images, generates personalized size charts, and accurately estimates fabric requirements. By combining design image analysis, user body data, and AI-based material calculation, the proposed system aims to improve garment fit while significantly reducing fabric waste in custom clothing production.

3.Research Problem and Questions

3.1 Main Research Problem

Despite advancements in artificial intelligence and digital tailoring technologies, there is currently a lack of integrated systems that utilize clothing design image analysis to generate personalized size charts and accurately estimate fabric requirements, resulting in poor garment fit and excessive fabric waste in custom clothing production.

3.2 Specific Research Questions

The proposed research work intends to solve the research problem by concentrating on these research questions:

1. What are some possible uses of artificial intelligence and image analysis algorithms to identify relevant design parameters from uploaded design pictures for clothing?

2. How can user-supplied body information be successfully combined with analysis of the design image in order to generate an accurate personalized garment size chart?

3. Through which mechanism can an AI-based system calculate the exact amount of fabric required for custom fashion production, based on design and individual body measurements?

4. How effective is the proposed AI-assisted fabric estimation technique in minimizing waste when compared with conventional fabric estimation techniques?

5. How effectively can the proposed web-based digital tailor system help increase the accuracy of clothing fits and the usage of sustainable fabrics?

4.Research Strategy and Methodology

4.1 Research Strategy

This research will use the methodology of **Design Science Research (DSR)**, which will also be appropriate for the problem-solving phase in the field of Information Systems and Software Engineering. According to the definition, the main point of Design Science Research is the design and evaluation of artifacts like software applications that provide solutions for problems existing in reality. The aim of the research will be the design and evaluation of the effectiveness of a working prototype based on the AI-enabled digital tailor system regarding its capability for creating a personalized size chart and minimizing fabric residues. The research will include the steps designed by Peffers et al.

4.2 Research Methodology Framework

The research will be guided by the Design Science Research Methodology (DSRM) framework. The phases are:

1. Problem Identification and Motivation
   Define the lack of integrated solutions that use design image analysis for size prediction and fabric estimation.
2. Define Objectives of a Solution
   Develop a system that can analyze images, generate sizes, and calculate fabric amounts accurately.
3. Design and Development
   Build the architecture and necessary AI models for image feature extraction and size prediction.
4. Demonstration
   Showcase the prototype with sample inputs and user interactions.
5. Evaluation
   Assess the system against traditional methods and measure fabric waste reduction.
6. Communication
   Document findings, results, and conclusions.

4.3 Data Collection Methods

To address the research problem of improving custom clothing fit and reducing fabric waste through an AI-assisted digital tailor system, this study adopts a mixed-method data collection approach. The data collection process combines secondary data for training the artificial intelligence-based image analysis model and primary data for developing and validating the fabric estimation and custom fitting logic.

This approach is necessary because the proposed system consists of two key components: an AI vision module that analyzes clothing design images and a logic-based module that calculates personalized size charts and fabric requirements**.**

4.3.1 Primary Data

Primary data will be collected firsthand to help build and test the system. It will include:

1. System evaluation data

- User interaction logs: Saved records of user inputs, chosen sizes, and system output from the prototype.
- User surveys: Online forms used to collect feedback about how well the clothes fit and how easy the system is to use.
- Observations: Notes made by the researcher while watching people use the system during usability tests, to find any problems.

2. Model validation data

- Fabric consumption data: Manually calculated fabric use for 10 standard garment designs with specific anthropometric inputs, validated against estimates provided by 3–5 professional tailors. These values serve as a ground-truth reference for statistical comparison against the system's predictions.
- Body measurement data: Sets of measurements for different body types (bust, waist, hip, height) used to test the custom-fit function.

4.3.2 Secondary Data

Secondary data is used to train the AI image analysis model. The **DeepFashion2 dataset** obtained from Kaggle is used for this purpose. This dataset contains a large number of labeled clothing images with garment categories and design features. These images help the system learn how to identify different types of clothing and design details from uploaded photos.

Dataset Link (to be cited in documentation):
https://www.kaggle.com/datasets/thusharanair/deepfashion2-original-with-dataframes

4.4 Data Preprocessing / Preparation

The system will be built in separate modules, as listed below:

1.User Interface (Frontend)

- Web pages for user registration/login, design image upload, body input, and chat assistant interface.

2.AI Processing Module (Backend)

- Image Analysis Subsystem: Uses deep learning models to extract design features from uploaded images.
- Size Prediction Subsystem: Combines extracted image features with user body data to generate personalized size charts.
- Material Estimation Subsystem: Calculates fabric usage based on size outputs and design attributes.

3.Database

- Stores user data, uploaded images, generated size profiles, and system logs.

4.Evaluation Module

- Implements comparison logic to measure fabric waste between traditional methods and the AI-based predictions.

The system will be developed as a web-based application using appropriate programming languages and frameworks (e.g., Python for AI, JavaScript/React for frontend, Node.js for backend).

4.5 Model Development / Implementation

A. Data Preprocessing

- Clean and organize image annotations from the DeepFashion2 dataset.
- Normalize user body input data.
- Balance data for image classes and garment types.

B. Feature Extraction (Computer Vision)

- Apply transfer learning using YOLOv8 to detect structural garment features including necklines, sleeve types, darts, and seam configurations from uploaded design images.
- Extract key landmarks and shape information relevant to sizing and material use.

C. Size Prediction Model

- Use supervised learning models to map extracted design features and anthropometric body metrics to predicted garment measurements, generating a dynamic 2D parametric drafting block per garment design.
- Validate model performance using cross-validation.

D. Material Estimation Logic

- Apply a Python parametric algorithm that derives a dynamic 2D drafting block from anthropometric inputs, using proportional scaling based on Winifred Aldrich's Metric Pattern Cutting standards and fabric-specific ease rules (including negative ease for stretch knit fabrics).
- Consider design complexity elements (necklines, sleeves, darts, collars) when computing pattern pieces and applying seam allowances.
- Compute exact fabric yardage using a 2D pattern nesting algorithm to minimize material waste, and generate a production-ready PDF drafting guide via the ReportLab library for tailor use.

4.6 Evaluation

The system will be evaluated on the following criteria:

4.6.1 Fit Accuracy

- Measure how closely predicted size charts match actual measurements.
- Comparison with standard size charts.

4.6.2 Fabric Waste Reduction

- Quantify fabric saved using the AI system versus traditional overcutting methods.
- Present percentage change in fabric usage.

4.6.3 Usability and User Satisfaction

- Analyze survey data and usability observations from participants.
- Likert scale feedback on ease of use and perceived accuracy.

4.7 Ethical considerations

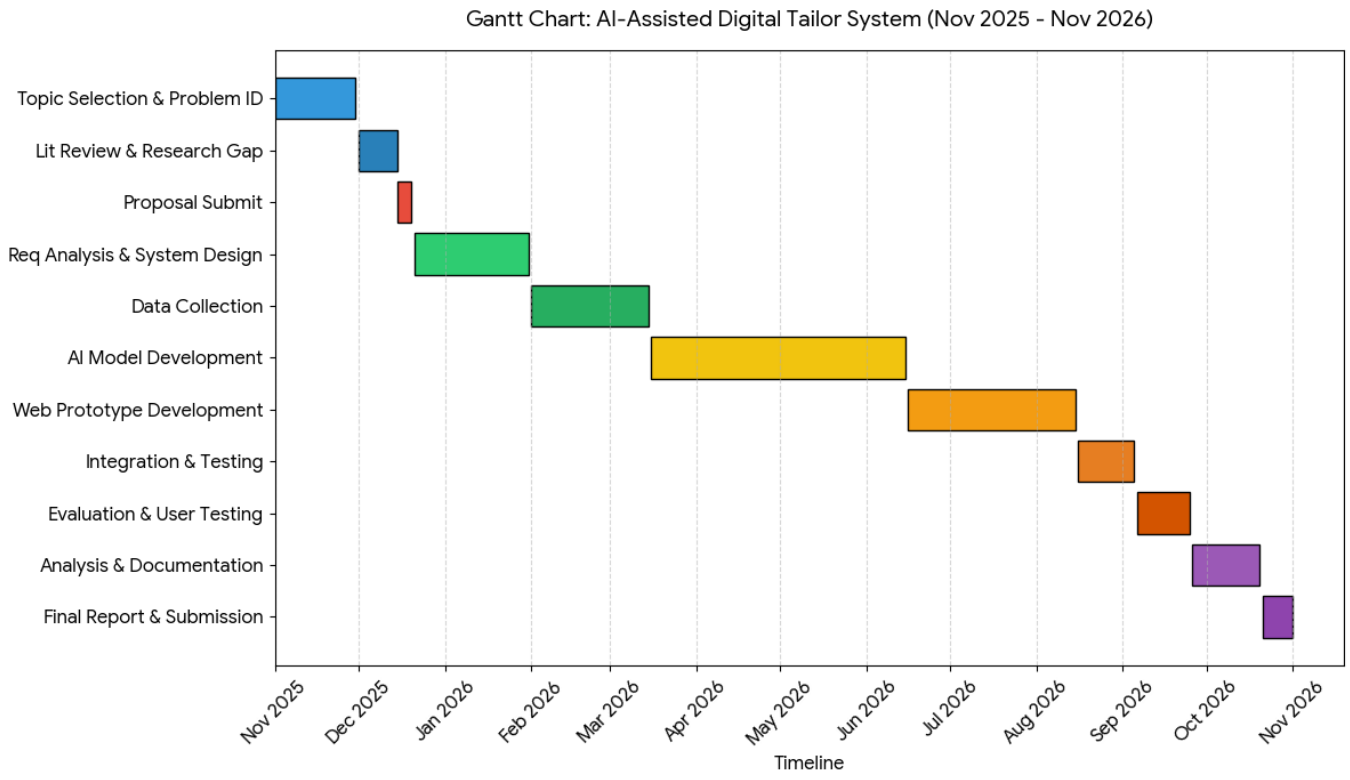
The research will ensure:

- User Consent: Participants will be informed and provide consent for surveys and observations.
- Data Privacy: Sensitive user data (e.g., body measurements) will be stored securely and used only for research purposes.
- Transparent Use of AI: Users will be informed about how their data is processed by the AI components.

4.8 Limitations of the study

- Dataset Bias: The DeepFashion2 dataset may not capture all body shapes and garment types.
- Generalization: AI model performance may vary for design images significantly different from training data.
- Scope Constraints: The system is scoped to ladies' casual garments – specifically blouses, T-shirts, and crop tops. Trousers, men's wear, and complex eveningwear are out of scope for this study.
- Image Ambiguity: Single front-view images may not capture full construction detail (e.g., back yoke, side seams), which may affect feature extraction accuracy. Additionally, user accuracy in entering body measurements will influence system output quality.

5.Timeline



Phase / Activity	Start Date	End Date	Key Deliverable
Topic Selection & Problem ID	Nov 2025	Nov 2025	Research Topic Approval
Literature Review & Research Gap	Dec 2025	Dec 2025	Literature Review Summary & Gap Analysis
Proposal Writing & Submission	Dec 2025	Dec 2025	Final Research Proposal Document
Requirement Analysis & System Design	Dec 2025	Feb 2026	SRS Document & System Architecture Diagrams
Data Collection (Primary & Secondary)	Feb 2026	Mar 2026	DeepFashion2 Dataset & Ground Truth Data
AI Model Development	Mar 2026	Jun 2026	Trained Computer Vision & Size Prediction
Web Prototype Development	Jun 2026	Aug 2026	Functional Frontend & Backend System
Integration & Testing	Aug 2026	Sep 2026	Fully Integrated System (AI + Web)
Evaluation & User Testing	Sep 2026	Oct 2026	Evaluation Report (Accuracy Metrics & User Feedback)
Analysis & Documentation	Oct 2026	Oct 2026	Complete Thesis Draft
Final Report & Submission	Nov 2026	Finalize	Final Dissertation & Viva Presentation

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